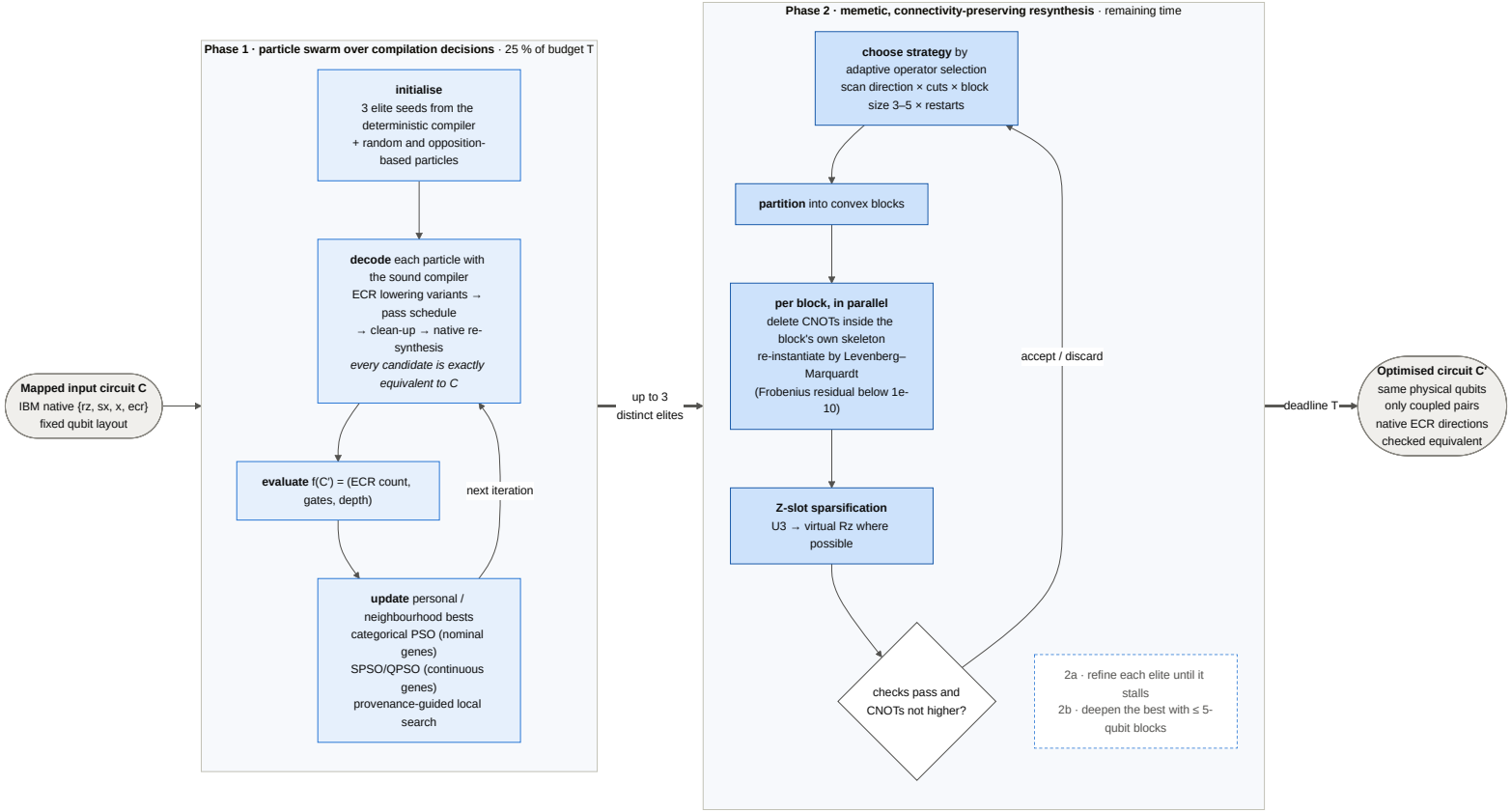

Mapped input circuit C
IBM native {rz, sx, x, ecr}
fixed qubit layout
Phase 1 · particle swarm over compilation decisions · 25 % of budget T
initialise
3 elite seeds from the deterministic compiler
+ random and opposition-based particles
decode each particle with the sound compiler
ECR lowering variants → pass schedule
→ clean-up → native re-synthesis
every candidate is exactly equivalent to C
evaluate f(C') = (ECR count, gates, depth)
next iteration
update personal / neighbourhood bests
categorical PSO (nominal genes)
SPSO/QPSO (continuous genes)
provenance-guided local search
up to 3 distinct elites
Phase 2 · memetic, connectivity-preserving resynthesis · remaining time
choose strategy by adaptive operator selection
scan direction × cuts × block size 3–5 × restarts
partition into convex blocks
per block, in parallel
delete CNOTs inside the block's own skeleton
re-instantiate by Levenberg–Marquardt
(Frobenius residual below 1e-10)
Z-slot sparsification
U3 → virtual Rz where possible
checks pass and CNOTs not higher?
accept / discard
2a · refine each elite until it stalls
2b · deepen the best with ≤ 5-qubit blocks
deadline T
Optimised circuit C'
same physical qubits
only coupled pairs
native ECR directions
checked equivalent